

Pancreas and Pancreatitis

Trypsin and its brakes (PRSS1, SPINK1, CTSC, CFTR, CLDN2), gallstones, triglycerides, calcium and pancreatic cancer

Sérgio F · 02-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. The trypsin's brakes are intact: SPINK1 without N34S (T/T), CTSC without the c.180T allele (C/C) and without the risk allele SPINK1 of GWAS. In **CLDN2-MORC4** (chromosome X) the only copy of Sérgio F has the protective allele C. This is the locus of pancreatitis with the most effect on men. The **CFTR R117H** is absent and the calcium sensor is also without risk variants (**CASR** R990G and A986S absent). In triglycerides there are no strong alleles: **APOA5** -1131 and S19W absent and **APOC3** without S2. In gallstones, **ABCG8 D19H** is absent. The pancreatitis loci load is in percentile 59.2. **ABO** suggests group O, which has less pancreatic cancer and less acute pancreatitis. The genomic score of pancreatic cancer is typical (percentile 45.0).

Unfavourable. The most relevant point is the biliary calculations: the genomic score is in percentile 88.8 and the 876 variant in the 71.6 (the loci load is in the 34.9, so the signal is moderate). It also has heterozygous UGT1A1 (Gilbert). In **PRSS1-PRSS2** it has 1 allele of more trypsinogen (T/C, OR ~1.4). The pancreatic cancer score of 22 variants is in percentile 83.1, with NR5A2, PDX1 and TP63 in risk homozygosity, although the genomic score is typical. Triglycerides are typical (68.2), with 1 risk allele in GCKR and TRIB1, and PNPLA3 I148M is heterozygous.

What does it mean? Hereditary pancreatitis and "genetic" (SPINK1, CTSC, CFTR, CLDN2) are favourable. The realistic risk comes from outside: gallstones, which are more likely genetically and also during a rapid loss of fat with the retatrutide, and triglycerides, that isotretinoin can rise. Zero alcohol already removes 2.a cause of pancreatitis.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Keyvariants** 30 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 26 of the chip, 3 imputed and 1 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 7 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 5 PGS Catalog scores calculated locally with 7.344.510 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MAIS PROVÁVEL/ALTO; ≤ 20 : MENOS PROVÁVEL/BAIXO; between 20 and 80: TÍPICO.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 204 loci, 159 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In all scores, high percentile = higher risk (in triglycerides, higher triglycerides).

Pancreatic cancer

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Pancreatic cancer	PGS000083 (Graff 2021, 22 variants)	21 22 (weighted coverage 97%)	83,1	MAIS PROVABLE	unfavourable
Pancreatic cancer (genomic)	PGS004693 (Hu 2024, AnnoPred, 6.35 million variants)	5.963.031 6.351.686 (weighted coverage 89%)	45,0	TIPIC	typical

Biliary calculations

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Biliary calculations	PGS001256 (Tanigawa 2022, 876 variants)	772 876 (weighted coverage 91%)	71,6	TIPIC	typical
Cololithiasis and cholecystitis	PGS002072 (Privé 2022, LDpred2, 428 thousand variants)	421.201 428.587 (weighted coverage 98%)	88,8	MAIS PROVABLE	unfavourable

Triglycerides

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Triglycerides	PGS002719 (Weissbrod 2022, SbayesR, 959 thousand variants)	959.485 981.472 (weighted coverage 97%)	68,2	TIPICS	typical

4. Key variants by topic

Variants with effect described in the literature. The judgment refers to the pancreas (pancreatitis and cancer) and its causes (biliary calculi, triglycerides and calcium).

Trypsin and pancreatic protection (pancreatitis)

Pancreatitis begins when trypsin activates inside the pancreas itself. There are genes that increase trypsinogen (PRSS1), activated trypsin (SPINK1, CTSC) and pancreatic juice flow genes (CFTR, CLDN2). CLDN2 is in the X chromosome, so a man has only one copy.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PRSS1-PRSS2	rs10273639		T/C	chip	C (1 copy/s)	unfavourable (light)	Cationic trypsinogen promoter: C allele increases PRSS1 expression and the risk of chronic and recurrent pancreatitis (OR ~1.4 by allele)	Strong
PRSS1-PRSS2	rs2855972		A/C	imputation DR2=0.98	A (1 copy/s)	unfavourable (light)	Second sign of the PRSS1-PRSS2 locus (chronic pancreatitis) [GWAS Catalog: Chronic pancreatitis (PheCode 577.2), allele C, beta 0.183 [0.15-0.22] unit decrease, p=5E-17, PMID 39024449] (the reported allele C reduces the trait; here the effect allele is the opposite, A)	Moderate
CTSC	rs497078		C/C	chip	T (0 copy/s)	favourable	c.180C>T (G60=): T allele reduces chymotrypsin C, the enzyme that degrades trypsin activated by mistake within the pancreas (OR ~2 in chronic pancreatitis)	Strong
SPINK1	rs17107315	N34S	T/T	chip	C (0 copy/s)	favourable	N34S: C allele (Ser34) is the classical variant of trypsin inhibitor (OR ~11 in idiopathic chronic pancreatitis; rare, ~1-2%)	Strong
SPINK1 (region)	rs148911734		C/C	imputation DR2=0.98	T (0 copy/s)	favourable	GWAS signal of SPINK1 region in acute and chronic pancreatitis [GWAS Catalog: Chronic pancreatitis (PheCode 577.2), allele C, beta 0.784 [0.71-0.86] unit decrease, p=2E-17, PMID 39024449] (the reported allele C reduces the trait; here the effect allele is the opposite, T)	Moderate
CLDN2-MORC4 (X)	rs12688220		C (X)	chip	T (0 copy/s)	favourable	Claudina-2 (junctions of pancreatic ducts): T allele increases the risk of chronic and acute pancreatitis (OR ~1.4; greater effect on men, who have only one copy of X)	Strong
CFTR	rs113993960	F508del	SNP not found in the data	SNP not found in the data	© 2011 Elsevier Ltd.	without data	F508del: the most common variant of cystic fibrosis; in carriers (1 copy) the risk of chronic pancreatitis rises ~3x	Strong
CFTR	rs78655421	R117H	G/G	chip	A (0 copy/s)	favourable	R117H: slight CFTR variant, associated with pancreatitis and male infertility due to absence of the deferens (rare) channels	Strong

Biliary calculus (main cause of acute pancreatitis)

Biliary stones are the most common cause of acute pancreatitis: a calculation that passes through the choledocus can block the pancreatic canal. ABCG5/ABCG8 releases cholesterol into the bile and UGT1A1 (Gilbert) increases bilirubin.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ABCG8	rs11887534	D19H	G/G	chip	C (0 copy/s)	favourable	D19H: C allele (His19) increases the secretion of cholesterol to bile and the risk of gallstones (OR ~2 by allele); also associated with acute pancreatitis	Strong
ABCG8	rs76866386		T/T	chip	C (0 copy/s)	favourable	ABCG8 signal directly associated with acute pancreatitis (via biliary stones) [GWAS Catalog: Acute pancreatitis, allele C, beta 0.376 [0.29-0.47] unit increase, p=2E-16, PMID 34594039]	Moderate
UGT1A1	rs6742078		G/T	chip	T (1 copy/s)	unfavourable (light)	UGT1A1*28 marker (Gilbert): more bilirubin and more pigmentary calculations (OR ~1.2)	Moderate
CYP7A1	rs3808607		G/G	chip	T (0 copy/s)	favourable	Promoter of the limiting enzyme of bile acids synthesis (-204A>C): relationship with bile cholesterol [GWAS Catalog: Total cholesterol levels (MTAG), T allele, beta 0.028 [0.021-0.035] unit increase, p=1E-14, PMID 37669986]	Weak
ABO	rs10901252		G/G	imputation DR2=0.99	C (0 copy/s)	favourable	Blood group ABO in acute pancreatitis (group O is protective) [GWAS Catalog: Acute pancreatitis, allele C, beta 0.140 [0.1-0.18] unit increase, p=2E-12, PMID 38491158]	Moderate

Triglycerides (metabolic cause of pancreatitis)

Very high triglycerides (above ~1000 mg/dL) are a classic cause of acute pancreatitis. These genes dictate the genetic basis of triglycerides.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
APOA5	rs662799		A/A	chip	G (0 copy/s)	favourable	-1131T>C: the risk allele rises to triglycerides (up to +20-30%); very high triglycerides are 3.the cause of acute pancreatitis [GWAS Catalog: Triglyceride levels, allele A, beta 0.292 [0.28-0.3] unit decrease, p=2E-525, PMID 34887591] (the allele A reported reduces the trait; here the effect allele is the opposite, G)	Strong
APOA5	rs3135506	S19W	G/G	chip	C (0 copy/s)	favourable	S19W: variant that rises the triglycerides [GWAS Catalog: Triglyceride levels, C allele, beta 0.243 [0.23-0.25] unit increase, p=4E-537, PMID 33462484]	Moderate
LPL	rs328	S447X	C/C	chip	G (0 copy/s)	neutral	S447X: G allele (Ter447) increases lipoprotein lipase activity and low triglycerides (protector)	Strong
GCKR	rs1260326	P446L	T/C	chip	T (1 copy/s)	unfavourable (light)	P446L: T allele rises triglycerides (and low fasting glucose)	Strong
APOC3	rs5128	SstI	C/C	chip	G (0 copy/s)	favourable	SstI (S2 = G allele): up the triglycerides and apoC-III	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
TRIB1	rs2954029		A/T	chip	A (1 copy/s)	unfavourable (light)	Strong locus of triglycerides [GWAS Catalog: Triglycerides, allele A, beta 0.083 unit increase, p=2E-108, PMID 29507422]	Moderate

Calcium and channels of pancreatic juice

High calcium activates trypsin within the pancreas. The calcium sensor (CASR) controls the calcium of blood and pancreatic juice.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CASR	rs1042636	R990G	A/A	chip	G (0 copy/s)	favourable	R990G: G allele increases the sensitivity of the calcium sensor; associated with chronic pancreatitis (OR ~2 in some studies)	Weak
CASR	rs1801725	A986S	G/G	chip	T (0 copy/s)	favourable	A986S: the risk allele rises to serum calcium (hypercalcemia = rare cause of pancreatitis) [GWAS Catalog: Calcium levels, T allele, beta 0.193 [0.19-0.2] unit increase, p=3E-595, PMID 33462484]	Moderate

Pancreatic cancer

Loci GWAS replicated from pancreatic adenocarcinoma. Blood group O is the most common known genetic factor of lower risk.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ABO	rs505922		T/T	chip	C (0 copy/s)	favourable	The C allele marks the non-O groups, with more pancreatic cancer (OR ~1.2 per allele); T/T suggests group O	Strong
CLPTM1L-TERT	rs401681		C/T	chip	T (1 copy/s)	unfavourable (light)	Pancreatic cancer locus 5p15.33 [GWAS Catalog: Pancreatic cancer, T allele, OR 1.19, p=9E-17, PMID 29422604]	Moderate
NR5A2	rs3790844		A/A	chip	A (2 copy/s)	unfavourable	Transcription factor of the acinar cells (pancreatitis and pancreatic cancer) [GWAS Catalog: Pancreatic cancer, allele A, OR 1.23, p=8E-16, PMID 29422604]	Moderate
PDX1-AS1	rs9581943		A/A	chip	A (2 copy/s)	unfavourable	Pancreatic development [GWAS Catalog: Pancreatic cancer, allele A, OR 1.15, p=5E-14, PMID 29422604]	Moderate
LINC00673	rs7214041		C/T	chip	T (1 copy/s)	unfavourable (light)	Pancreatic cancer locus 17q25.1 [GWAS Catalog: Pancreatic cancer, T allele, OR 1.25, p=9E-15, PMID 29422604]	Moderate
TP63	rs9854771		G/G	chip	G (2 copy/s)	unfavourable	Pancreatic cancer locus 3q28 [GWAS Catalog: Pancreatic cancer, G allele, OR 1.12, p=2E-8, PMID 26098869]	Weak
ZNRF3	rs16986825		C/C	chip	T (0 copy/s)	favourable	By Wnt (pancreatic cancer) [GWAS Catalog: Pancreatic cancer, T allele, OR 1.15, p=1E-8, PMID 29422604]	Weak

Fat and function of the pancreas

Fat in the pancreas follows fat in the liver (PNPLA3). The locus CTRB1-CTRB2 binds digestive enzymes to diabetes.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PNPLA3	rs738409	I148M	C/G	chip	G (1 copy/s)	unfavourable (light)	I148M: the G allele accumulates fat in the liver and is also associated with more fat in the pancreas and more severe pancreatitis	Weak
CTRB1-CTRB2	rs7202877		T/T	chip	G (0 copy/s)	favourable	Chemotrypsinogen B (digestive enzyme): the G allele is associated with more 1 type diabetes (OR ~1.28); the locus also touches the response to incretins [GWAS Catalog: Type 1 diabetes, allele G, OR 1.28, p=3E-15, PMID 19430480]	Moderate

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of each disease, with each allele oriented towards the risk, compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Pancreatitis (acute and chronic)	37 (21)	8 de 42	21	59,2	typical
Biliary calculations	125 (105)	92 de 210	105	34,9	typical
Pancreatic cancer	42 (33)	31 de 66	33	60,4	typical

6. Favourable

- **SPINK1 N34S absent** (T/T) and CTSC c.180C>T absent (C/C): the two main brakes of trypsin are normal. SPINK1 of GWAS without the risk allele. **Strong.**
- **CLDN2-MORC4 with the protective allele** (chromosome X, hemizygote C (X)). It is the strongest locus of pancreatitis in men. **Strong.**
- **CFTR R117H absent**; CASR R990G and A986S absent. **Moderate.**
- **Triglycerides without strong alleles**: APOA5 -1131 (A/A) and S19W (G/G) absent; APOC3 SstI without S2. **Strong.**
- **ABCG8 D19H absent** (G/G), without the other ABCG8 signal of acute pancreatitis, and CYP7A1 without the risk allele. **Moderate.**
- **Load of pancreatitis loci in percentile 59.2** (typical). **Moderate.**
- **Probable group O** (ABO rs505922 T/T): less pancreatic cancer and less acute pancreatitis. **Moderate.**
- **Pancreatic cancer: typical genomic score** (percentile 45.0, 6 million variants); loci load in 60.4. ZNRF3 without risk. **Moderate.**
- **CTRB1-CTRB2 T/T**: without the allele associated with 1 type diabetes. **Weak.**

7. Unfavourable

- **Biliary calculations:** percentile 88.8 in the genomic score (428 thousand variants) and 71.6 in the 876 variant. The loci load is in 34.9, so the signal is mixed. Calculations are the main cause of acute pancreatitis. **Moderate.**
- **UGT1A1 rs6742078 G/T** (Gilbert heterozygote): more bilirubin and pigmentary calculations. **Weak moderated.**
- **PRSS1-PRSS2:** rs10273639 T/C and rs2855972 A/C, with 1 risk allele each (more trypsinogen). **Moderate.**
- **Pancreatic cancer (22 variant score):** percentile 83.1, with NR5A2 A/A, PDX1-AS1 A/A and TP63 G/G in risk homozygosity, plus CLPTM1L and LINC00673 heterozygous. The genomic score is typical, so the set is slight. **Weak moderated.**
- **Triglycerides:** percentile 68.2 (typical), with GCKR P446L T/C and TRIB1 A/T heterozygotes. **Weak.**
- **PNPLA3 I148M C/G:** more fat in the liver and possibly in the pancreas. **Weak.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Endoplasmic Reticulum Disorder	Acinar cells: PRSS1 with 1 allele of more trypsinogen (more double enzyme load); SPINK1 and CTSC normal	Neutral
Lysosome dysfunction	Early activation of trypsin by cathepsin B in lysosomes is blocked by normal SPINK1 and CTSC	Favourable
Chronic Inflammation	Pancreatitis: loci load in percentile 59.2; CLDN2 protector	Favourable
Intercellular Communication	Claudina-2 (junctions of pancreatic ducts) with the protective allele	Favourable
Metabolic Homeostasis Decline	Triglycerides in percentile 68.2; without risk APOA5/APOC3; GSKR and TRIB1 heterozygotes	Neutral
Dysfunction of adipose tissue	PNPLA3 I148M heterozygous (ectopic fat in liver and pancreas)	Unfavourable (mild)
Hyperglycemia and Intolerantness to Glucose	Endocrine pancreas: CTRB1-CTRB2 without the risk allele; PDX1 in risk homozygosity (locus of pancreas development)	Mixed
Somatic Mutation Accumulation	Pancreatic cancer: 22 score variants in percentile 83.1, genomic score in the 45.0; group O probable	Neutral to unfavourable (mild)

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Base abdominal ultrasound** (vesicle, bile ducts and liver), especially during retatrutide. The genomic score of calculations is in the ~the 89th percentile and a rapid loss of body fat increases the formation of calculations. *Suggestion (not applied).*
2. **Ursodeoxycholic acid during the rapid loss of body fat:** prevents the formation of calculations in accelerated loss (evidence in bariatric surgery). Discuss with the doctor if the ultrasound shows biliary mud or if the loss is rapid. *Suggestion (not applied).*
3. **Do not skip meals with fat:** with the retatrutide and feeding window of 10–12 h, include olive oil in each main meal (40–50 mL/day, as in the plane). This keeps the vesicle contracting and reduces biliary stasis. *Suggestion (not applied).*
4. **Pancreatitis alarm signs** (strong pain in the epigastrium that radiates to the back, with vomiting): stop the retatrutide and ask for lipase on the day itself. Routine lipase without symptoms is not indicated. *Suggestion (not applied).*
5. **Fasting triglycerides in isotretinoin controls** (that rises them). Omega-3 of 3.2–4 g/day, fructose added $\approx$ 0 and zero alcohol from the plane are the main levers. *Suggestion (not applied).*
6. **Maximum fiber** (psyllium + PHGG; the more, the better): low bile cholesterol and triglycerides. LDL and ApoB as low as possible with ezetimiba, which also lowers cholesterol absorption and bile cholesterol. *Suggestion (not applied).*
7. **Calcium:** do not add calcium in supplement to Ca-AKG without need. Vitamin D follows the plan. *Suggestion (not applied).*
8. **Pancreatic cancer:** the genes do not justify screening by image, because the genomic score is typical and there are no known hereditary variants. A new diabetes without explanation from the 50 years should make us think about the pancreas. *Suggestion (not applied).*
9. **Zero alcohol and no smoking** (already underway): are the two strongest preventable causes of chronic pancreatitis and pancreatic cancer. *Suggestion (not applied).*

Not recommended: alcohol; repeated prolonged fasting without fat during rapid loss of body fat.

10. Limitations

- A genotyping chip with imputation does not read most of the rare variants of PRSS1 (R122H, N29I), CPA1, CEL (including the hybrid CEL-HYB) or the rare variants of CFTR. The **CFTR F508del** (3 bases) is not in the data, so the state of cystic fibrosis carrier is not evaluated in this report (see the carrier status report).
- There are no polygenic scores of pancreatitis in the PGS Catalog with sufficient quality. Pancreatitis is evaluated by the key genes and by the loci load of GWAS Catalog.
- Biliary calculi scores are based on health records. The 22 variant of pancreatic cancer has little predictive capacity, and the genome of 6 million variants is more stable.
- The load of loci GWAS sums SNPs in linkage imbalance. It is a descriptive complement and does not replace the polygenic scores.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. The variants of the X chromosome (CLDN2-MORC4) come only from the chip, in hemizygosity. No genetic data came out of the analysis computer.
- **Alleles and strand:** The alleles of each location are read from the Imputed VCF. The published effect allele is compared with those alleles and, if necessary, converted to the complementary strand (only in non-palindromic SNPs). In palindromics (ABCG8 D19H, LPL S447X, APOA5 S19W, APOC3 SstI, PNPLA3 I148M) the strand + is assumed. In the loci of GWAS Catalog with protective effect, the allele presented is that of risk.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog: PGS000083 (Graff 2021) and PGS004693 (Hu 2024) for pancreatic cancer PGS001256 (Tanigawa 2022) and PGS002072 (Privé 2022) for gallstones PGS002719 (Weissbrod 2022) for triglycerides Score = Δ coefficient $\times$ dose of effect allele. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog (associations with $p \leq 5 \times 10^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** GWAS Catalog, ClinVar, dbSNP/Ensembl and reference literature: Whitcomb 2012 and Rosendahl 2018 (PRSS1-PRSS2 and CLDN2), Witt 2000 (SPINK1), Masson 2022 (CTRC), Buch 2007 (ABCG8), Klein 2018 (pancreas cancer). The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Trypsin and pancreatic protection (pancreatitis)	PRSS1-PRSS2	rs10273639	T/C	chip	C	1	strand +	1 copy of risk allele
Trypsin and pancreatic protection (pancreatitis)	PRSS1-PRSS2	rs2855972	A/C	imputation DR2=0.98	A	1	strand +	1 copy of risk allele
Trypsin and pancreatic protection (pancreatitis)	CTRC	rs497078	C/C	chip	T	0	strand +	without the risk allele
Trypsin and pancreatic protection (pancreatitis)	SPINK1	rs17107315	T/T	chip	C	0	strand +	without the risk allele
Trypsin and pancreatic protection (pancreatitis)	SPINK1 (region)	rs148911734	C/C	imputation DR2=0.98	T	0	strand +	without the risk allele
Trypsin and pancreatic protection (pancreatitis)	CLDN2-MORC4 (X)	rs12688220	C (X)	chip	T	0	strand +; hemizigote (chromosoma X)	without the risk allele (X, hemizigote)
Trypsin and pancreatic protection (pancreatitis)	CFTR	rs113993960	SNP not found in the data	SNP not found in the data	D			SNP not found in the data
Trypsin and pancreatic protection (pancreatitis)	CFTR	rs78655421	G/G	chip	A	0	strand + (published alleles; absent site of VCF)	without the risk allele
Biliary calculus (main cause of acute pancreatitis)	ABCG8	rs11887534	G/G	chip	C	0	strand + (palindromic; assumed +)	without the risk allele
Biliary calculus (main cause of acute pancreatitis)	ABCG8	rs76866386	T/T	chip	C	0	strand +	without the risk allele
Biliary calculus (main cause of acute pancreatitis)	UGT1A1	rs6742078	G/T	chip	T	1	strand +	1 copy of risk allele
Biliary calculus (main cause of acute pancreatitis)	CYP7A1	rs3808607	G/G	chip	T	0	strand +	without the risk allele
Biliary calculus (main cause of acute pancreatitis)	ABO	rs10901252	G/G	imputation DR2=0.99	C	0	strand + (palindromic; assumed +)	without the risk allele
Triglycerides (metabolic cause of pancreatitis)	APOA5	rs662799	A/A	chip	G	0	strand +	without the risk allele
Triglycerides (metabolic cause of pancreatitis)	APOA5	rs3135506	G/G	chip	C	0	strand + (palindromic; assumed +)	without the risk allele

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Triglycerides (metabolic cause of pancreatitis)	LPL	rs328	C/C	chip	G	0	strand + (palindromic; assumed +)	without the protective allele (common genotype)
Triglycerides (metabolic cause of pancreatitis)	GCKR	rs1260326	T/C	chip	T	1	strand +	1 copy of risk allele
Triglycerides (metabolic cause of pancreatitis)	APOC3	rs5128	C/C	chip	G	0	strand + (palindromic; assumed +)	without the risk allele
Triglycerides (metabolic cause of pancreatitis)	TRIB1	rs2954029	A/T	chip	A	1	strand + (palindromic; assumed +)	1 copy of risk allele
Calcium and channels of pancreatic juice	CASR	rs1042636	A/A	chip	G	0	strand +	without the risk allele
Calcium and channels of pancreatic juice	CASR	rs1801725	G/G	chip	T	0	strand +	without the risk allele
Pancreatic cancer	ABO	rs505922	T/T	chip	C	0	strand +	without the risk allele
Pancreatic cancer	CLPTM1L-TERT	rs401681	C/T	chip	T	1	strand +	1 copy of risk allele
Pancreatic cancer	NR5A2	rs3790844	A/A	chip	A	2	strand +	2 risk allele copies
Pancreatic cancer	PDX1-AS1	rs9581943	A/A	chip	A	2	strand +	2 risk allele copies
Pancreatic cancer	LINC00673	rs7214041	C/T	chip	T	1	strand +	1 copy of risk allele
Pancreatic cancer	TP63	rs9854771	G/G	chip	G	2	strand +	2 risk allele copies
Pancreatic cancer	ZNRF3	rs16986825	C/C	chip	T	0	strand +	without the risk allele
Fat and function of the pancreas	PNPLA3	rs738409	C/G	chip	G	1	strand + (palindromic; assumed +)	1 copy of risk allele
Fat and function of the pancreas	CTRB1-CTRB2	rs7202877	T/T	chip	G	0	strand +	without the risk allele